

Design optimization and numerical investigation of multi tube metal hydride reactor for large capacity hydrogen storage application

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ABSTRACT

The efficient operation of a metal hydride based hydrogen storage system primarily depends on the configuration of the storage reactor. Both the total weight and effective performance are crucial for large capacity hydrogen storage systems. Therefore, the present work is centered around the development of a lightweight, compact, and efficient MH reactor capable of accommodating a large hydrogen inventory. In this perspective, a disc finned multi tube reactor (DFMTR) design is proposed, and its behaviour is numerically analyzed. Five reactor configurations are studied using 3D COMSOL modeling to visualize the impact of various fin structures during the absorption and desorption half cycles. The DFMTR model shows remarkably higher performance by saving 38% and 31% time to store and release hydrogen compared to the baseline reactor. Further, the design methodology is discussed, and the geometrical and operational parameters are optimized to achieve the excellent performance of the proposed reactor. Furthermore, the portable design approach for scaling up the capacity of the storage system is discussed and recommended. Moreover, the proposed reactor design is compared with recent studies of the literature, and found that this reactor saved 22.1%, 50%, 52.1%, and 8.7% of time to store 90% hydrogen compared with tapered finned multi tube reactor, tube bundle reactor, radial finned reactor, and pin fin reactor, respectively. The authors believe that the proposed large capacity reactor has the potential to deliver outstanding performance, and this research will be instrumental in the construction of compact, effective metal hydride reactors with significant hydrogen storage capacities.

Introduction

It is imperative to expand the utilization of renewable resources on a larger scale by considering future energy requirements and the increasing carbon emissions in the world [1]. To accomplish the objectives of Paris Agreement, world emissions must be reduced by 30% by 2030, as reported by the Bloomberg Energy Outlook 2021 [2]. In this regard, finding an energy carrier that solves the intermittent issue of renewables and satisfies future energy needs without any negative effects on the environment is crucial [3]. Hydrogen is gaining popularity as a source in wide applications because it is a cleaner, abundant, flexible, and efficient energy carrier. The hydrogen energy economy includes hydrogen generation, storage, transportation, and its end use. Further, hydrogen offers an appealing choice since, whether it's burned for heat generation or combined with air in fuel cells to generate electricity, the only resulting by product is water. Nowadays, hydrogen can be generated using commercial electrolyzers, which can provide

hydrogen up to a pressure of 30 bar [4]. However, the development of safer, more efficient, and more affordable hydrogen storage technology is crucial. The traditional methods of storing green hydrogen, such as cryogenic liquid and compressed gas in high pressure vessels, consume up to 30% of the stored hydrogen energy content [5]. In recent years, sorption hydrogen storage using metal hydrides has gained attention due to its higher energy density ($\sim 140 \text{ MJ.kg}^{-1}$) [6]. In general, the MH systems are thermally driven, in which exothermic and endothermic chemical reactions occur during the absorption and desorption half cycles. The rate of heat transfer should be high enough to permit faster and continuous hydrogen gas storage and discharge. The metal hydrides (MHs) have poor thermal conductivity ($\sim 1 \text{ W.m}^{-1}\text{K}^{-1}$), they require an effective heat transfer system [7–9]. Hence, for the efficient performance of a metal hydride system, the heat transfer rate is a crucial factor. Metal hydride hydrogen storage systems (MHSS) can be integrated with electrolyzers to store the produced hydrogen under atmospheric conditions. Moreover, this thermally driven system requires only waste heat (in the range of 40–100 °C) for hydrogen retrieval.

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Nomenclature		Greek letters	
C_R	rate constant (s^{-1})	γ_H	hysteresis factor
C_p	specific heat ($J.kg^{-1}K^{-1}$)	ε_p	porosity
A_e	activation energy ($J.mol^{-1}$)	α_s, α_e	slope constant
$\bar{h}$	heat transfer coefficient ($W.m^{-2}K^{-1}$)	Nu	average nusselt number
k	thermal conductivity ($W.m^{-1}K^{-1}$)	ρ	density ($kg.m^{-3}$)
M	molecular weight ($kg.mol^{-1}$)	Subscript	
R_u	universal gas constant ($J.kg^{-1}K^{-1}$)	s	solid
T	temperature (K)	f	gas
wt.%	hydrogen storage capacity	i	initial
r, θ, z	axis coordinates	Acronyms	
ΔH	reaction enthalpy ($J.mol^{-1}$)	MH	Metal hydride
ΔS	reaction entropy ($J.mol^{-1}K^{-1}$)	HTF	Heat transfer fluid
x	concentration factor	OCT	Outer cooling jacket
$\vec{u}_r, \vec{u}_z$	gas velocities ($m.s^{-1}$)	lpm	litres per minute

Over the years, various techniques, including innovative reactor geometries and the incorporation of heat enhanceive fillers, have been reported in the literature to boost heat transfer within MH reactors. These techniques include the incorporation of different heat transfer tubes and coils [10–21], diverse extended surfaces [22–29], expanded natural graphite (ENG) compacts [30–32], and metallic fillers [33–35] within the MH reactor. Gkanas et al. [21] conducted a numerical study on the optimization of the rectangular type reactor for hydrogen storage applications. They studied the impact of varied bed thickness and tube positions and found the optimized thickness between two cooling or heating surfaces as 10.4 mm under a heat transfer coefficient of 2000 $W.m^{-2}K^{-1}$. Raju et al. [12] carried out the experimental study on 4 kg of $LaNi_{4.7}Al_{0.3}$ powder inside a 55 cooling tube reactor. They found that 46.2 g of hydrogen was absorbed by the alloy in 5 min at inlet pressure and heat transfer fluid (HTF) temperature of 20 bar and 293 K, respectively. Anbarasu et al. [36] carried out an experimental study on embedded cooling tube (ECT) reactors containing 2.75 kg of $LmNi_{4.91}Sn_{0.15}$ alloy. They compared two reactors (36 ECT reactor and 60 ECT reactor) and found that increasing the cooling tubes from 36 to 60 reduced the time taken by the alloy to reach 1.04 wt.% from 480 to 360 s under 25 bar supply pressure. Chung et al. [37] conducted an experimental study on reactors with or without heat pipes. They found the absorption time was dropped from 2100 s to 900 s by using a heat pipe as compared to not using a heat pipe under a gas pressure of 10 bar and HTF temperature of 293 K. Mellouli et al. [38] investigated the single helical tube reactor that contained 1 kg of $LaNi_5$ powder and found that the reactor took 25 min to reach 90% of its hydrogen capacity at 10 bar supply pressure. Also, the two helical tubes took 8.33 min, and one helical tube with fins also took the same time (8.33 min) to reach 90% saturation under similar operating conditions. Muthukumar et al. [15] formulated a two dimensional model filled with $MmNi_{4.6}Al_{0.4}$ and found that increasing the cooling tubes from 12 to 20 reduces the time to reach 90% saturation from 296 s to 155 s, respectively, under a gas pressure of 20 bar and an HTF temperature of 30 °C.

Numerous studies in the literature have suggested that a combination of tubes and fins can significantly enhance the effectiveness of the MH reactor. Parida and Muthukumar [23] numerically investigated the impact of dissimilar fin types, namely longitudinal, transverse, and spiral, on reactor performance. They concluded that longitudinal and transverse fins indicated slightly better performance during absorption and desorption processes, respectively. Further, the designed reactor was able to deliver hydrogen at 2.27 lpm till 2000 s at a discharge pressure of 10 bar. Askri et al. [24] investigated the different heat transfer systems. The reactor without any thermal management, took 2700 s to reach 90% of its maximum capacity under the 8 bar gas

pressure and HTF temperature of 293 K. Also, the absorption time was reduced to 2080 s, 980 s, and 460 s by adding the exterior fins, cooling tube, and cooling tube with internal fins, respectively. Singh et al. [25] conducted an experimental study on the MH reactor containing a U tubes with radial fins. They found that the reactor took 490 s to reach 1.2 wt.% saturation at 15 bar supply pressure and 15 °C HTF temperature. Dhaou et al. [26] carried out an experimental study on reactors containing 1 kg of $LaNi_5$ alloy. Two reactor designs, namely a simple reactor with an external jacket and a reactor with a spiral finned tube, were compared. It was observed that the absorption time dropped by 72.9% by using the spiral finned tube within the reactor. Ma et al. [28] conducted an optimization study on a multi tube finned reactor. The impact of variations in fin numbers, thickness, and radius was studied, and it was concluded that the optimized reactor design took 1720 s for complete absorption at 1 MPa gas pressure. Chandra et al. [29] numerically investigated a multi tube reactor attached to conical fins. They compared the effects of varying the tube and fins on reactor performance and concluded that the multi tube reactor having 6 tubes attached to 19 fins absorbed 80% and 90% of hydrogen in 290 s and 375 s at a gas pressure of 15 bar and a HTF temperature of 20 °C. Recently, a few studies reported advanced heat exchange structures. Wang et al. [39] conducted an optimization study on a novel disc mini channel reactor containing 1.56 kg of $LaNi_5$ powder. Five reactor designs were studied and it was found that this novel reactor saved 89% and 93% of the time during the absorption and desorption processes compared to a straight tube reactor. Bei et al. [40] numerically investigated a novel reactor attached to tree shaped fins. The results revealed that this reactor design saves 20.7% of absorption time compared to a radial finned reactor.

In the realm of large-capacity hydrogen storage, the shell and tube geometry have been extensively employed by numerous researchers. Raju et al. [41] developed the design methodology to optimize the cooling tubes for 50 kg capacity ECT reactors. Jana et al. [42] investigated a tube bundle reactor (TBR) with a 25 kg alloy capacity. They compared the performance of the TBR with the ECTR and found both reactors took 10 min to reach 1.24 wt.% capacity. Also, the weight ratio of the TBR was 1.17 times higher. The details of recently reported large capacity reactors for hydrogen storage are listed in Table 1.

According to the literature, a hydrogen storage system's reactor design is crucial to its effective operation. There have been numerous studies reported related to the design of the heat exchanger for hydrogen storage systems. However, it's worth noting that the majority of these studies were conducted on a laboratory scale. For large capacity stationary hydrogen storage, both the overall reactor mass and performance are of prime importance. The reported studies on large capacity

Table 1

Recent studies on large-capacity hydrogen storage reactors reported in the literature.

Author, Year [Ref.]	Alloy capacity	Reactor type	Conclusions
Kumar et al. 2019 [43]	40 kg of $\text{LaNi}_{4.7}\text{Al}_{0.3}$	ECT reactor	99 tubes embedded inside the reactor. 424.8 g of hydrogen was absorbed by the reactor in 1253 s at 10 bar supply pressure. 394.7 g of hydrogen was desorbed by the reactor in 3150 s at 70 °C HTF temperature.
Karmakar et al. 2021 [44]	10 kg of LaNi_5	ECT reactor	Four tubes with an external jacket were fixed inside the reactor. Under the supply pressure of 25 bar the MH bed absorbed 118.3 g of hydrogen in the 1620 s. 115 g of hydrogen was desorbed in 2700 s at 70 °C HTF temperature.
Mallik and Sharma 2021 [45]	50 kg of LaNi_5	ECT reactor	Eight tubes are attached with longitudinal fins. The reactor reached 1.21 wt.% capacity in 700 s at 20 bar supply pressure and 20 °C HTF temperature.
Afzal et al. 2021 [46]	47.5 kg of $\text{La}_{0.9}\text{Ce}_{0.1}\text{Ni}_5$	Hexagonal honeycomb reactor	The reactor reached the saturation capacity of 0.9 in 90 min at 25 bar supply pressure and 25 °C HTF temperature. The reactor desorbed 90% of hydrogen in 166.67 min at a HTF temperature of 50 °C.
Prasad and Muthukumar 2021 [47]	9 kg of LaNi_5	Annular reactor	The reactor consists of a single central tube embedded with 91 radial fins with an external jacket. The reactor was able to store 90% of hydrogen capacity (104 g) in 640 s at 20 bar supply pressure. 104 g of hydrogen was released in 1372 s under 60 °C HTF temperature.
Jana et al. 2022 [48]	21.5 kg of $\text{La}_{0.9}\text{Ce}_{0.1}\text{Ni}_5$	ECT reactor	The reactor absorbed 282.94 g of hydrogen in 912 s at 30 bar supply pressure. 282.8 g of hydrogen was released from the reactor in 4860 s under 40 °C HTF temperature.

reactors are limited and typically use the shell and tube geometry [43–45]. In general, large-capacity reactors are extended versions of lab-scale reactors that can accommodate a large hydrogen inventory. However, these structures involve bulkiness, longer hydrogen storage or discharge times, and manufacturing challenges. Hence, there is a crucial need for a compact, lightweight, cost effective, and efficient MH reactor design to accommodate large hydrogen inventories in a very fast way.

The present study proposes a lightweight multi tube disc finned reactor that offers excellent performance, particularly with a higher

mass ratio (ratio of alloy mass to reactor mass). The possible way of improving the reactor's hydrogen storage performance with the minimum reactor mass is through the utilization of extended surfaces inside the reactor. Using the extended surfaces inside the reactor increases the surface area of heat transfer, which helps in faster removal of the exothermic heat via HTF and improves the reactor's hydrogen storage performance. However, a detailed comparison between the different fin structures is required to maximize the reactor's performance. Therefore, the present study is focused on the detailed study of different fin structures within the reactor. Five reactor structures are analyzed during the absorption and desorption half cycles in Section 3.1. Also, the reactor design methodology, considering both overall weight and performance, has been discussed. The strategy for fin distribution to achieve the optimum thickness and fin numbers is described in Section 3.2. Furthermore, the performance of optimized reactor designs under various operating conditions is analyzed in Section 3.3. Besides this, Section 3.4 deals with the scale-up methodology and the outcomes of the DFMTR model compared with the recent studies reported in the literature.

Reactor design and numerical implementation

Design description

The prime objective of this study is to design a large capacity MH reactor targeting stationary hydrogen storage applications. The inspiration behind this work is to develop a design approach that effectively optimizes the reactor's heat transfer performance by considering crucial factors including size, weight, strength, and cost. The present study proposes the shell and tube type cylindrical geometry, which consists of multi tubes attached with disc fins surrounded by metallic powder inside a shell with an external jacket, as shown in Fig. 1b. The shell diameter of 88.9 mm with a thickness of 3.2 mm is chosen to endure a maximum pressure of 95 bar, and the reactor length of 1 m is adopted to accommodate ~20 kg of the alloy. The geometrical details of the reactor are represented in Fig. 1c. The HTF tube diameter of 12.7 mm with a thickness of 1.65 mm is optimally chosen by considering the heat transfer area in mind. To ensure the compactness and light weight of the MH reactor, a total of six HTF tubes is selected by considering the minimum mass ratio of 1.5, given in Table 2.

The reactor mass encompasses the shell, HTF tubes, filter, hydrogen tube, external jacket, end plates, and extended surfaces. The front and back end plates having a thickness of 10 mm are employed to seal the reactor. A 6 mm thickness is provided between the shell and external jacket to facilitate the HTF flow. A porous filter having an outer diameter of 9.5 mm is fitted at the center of the reactor for hydrogen exchange. The optimized design consists of 94-disc type extended surfaces, each with a thickness of 0.5 mm and a pitch of 10.5 mm, have been affixed along the axis to improve heat transfer within the reactor. Since, heat transfer is the crucial factor in MH systems [8], different MH reactor designs (similar to the proposed reactor, the only difference is the type of extended surface used within the reactor) are analyzed to determine the most efficient reactor design, as shown in Fig. 1a.

Material selection

In this study, LaNi_5 is chosen as the metal hydride alloy to facilitate a more straightforward comparison with existing literature. Additionally, this alloy demonstrates lower hysteresis in comparison to other AB_5 alloys. Table 3 presents the thermochemical properties of the alloy and hydrogen gas used for simulation. Freely accessible water having moderate thermal conductivity and exceptional stability is selected as HTF. For medium pressure applications, readily accessible ASTM (A-321) standard seamless tubes composed of SS-316 material are used to design the reactor. Copper is used as the material for fins because of its higher thermal conductivity and outstanding corrosion resistant property.

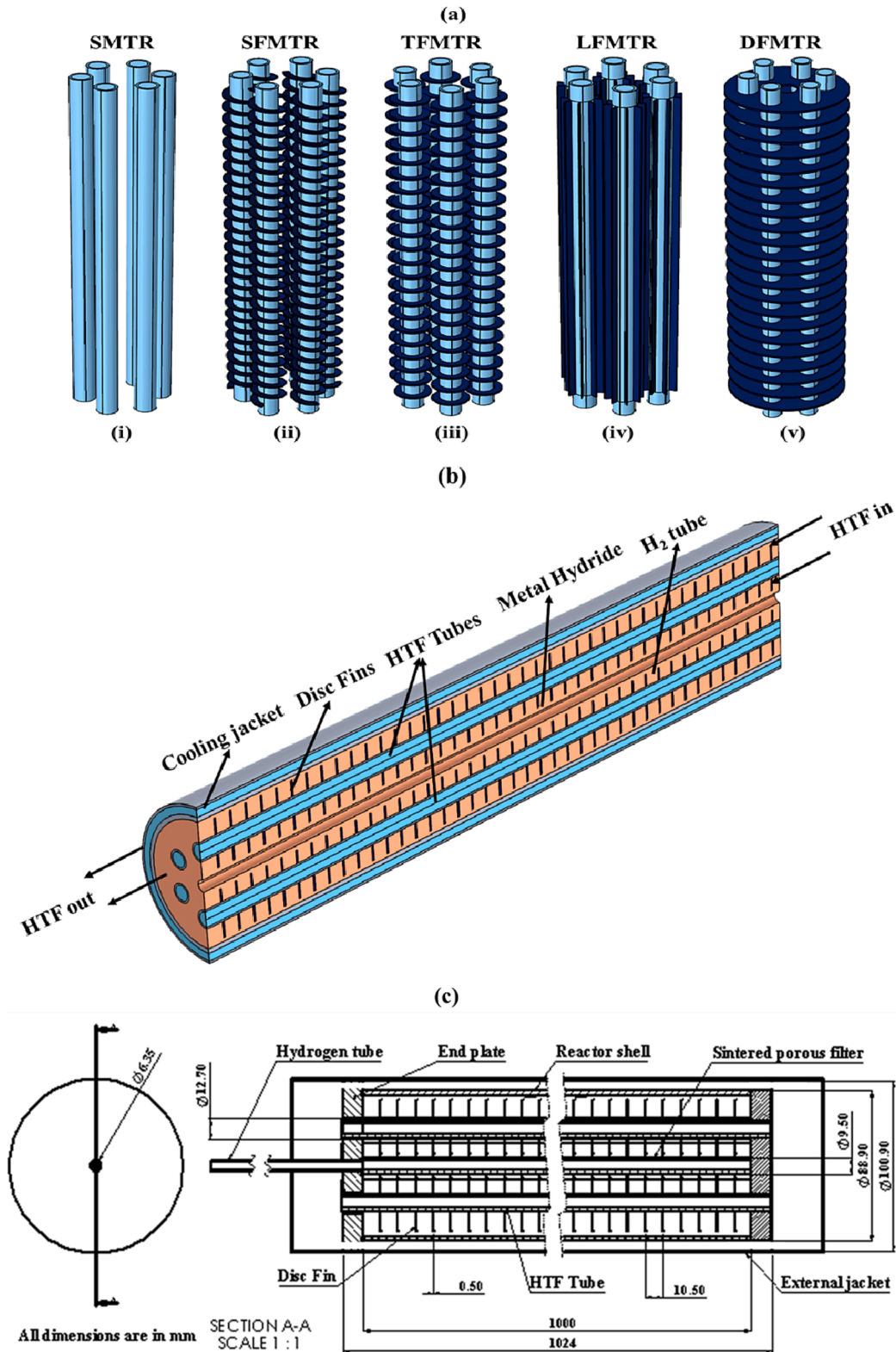


Fig. 1. (a) Schematic of various heat exchange systems, (b) Isomeric view, and (c) cross-sectional view of multi-tube reactor with disc fins (DFMR model).

Problem formulation

This study solves a 3D numerical model using COMSOL 5.3a for the absorption or desorption half cycles. These half cycles include the combined heat and mass transfer aspects.

Therefore, the porous domain adheres to the mass, momentum, and

energy conservation equations. Moreover, there is no specific trend for the porosity variations with hydrogen concentration, and the permeability has no significant effect in the MH systems. Therefore, the authors assumed these parameters to be constant. Further, the interface connecting the porous bed and HTF tubes is associated with the heat flux condition instead of direct flow to reduce the computational workload.

Table 2

Criteria for selection of the number of HTF tubes.

S. No	N_t	M_{MH}	M_R	$V/A_{s,t}$	M_{MH}/M_R
1	3	21.01	11.38	14.16	1.84
2	4	20.48	11.87	10.62	1.72
3	5	19.95	12.38	8.50	1.61
4	6	19.42	12.85	7.08	1.51
5	7	18.89	13.34	6.07	1.41

Table 3

Thermochemical properties of alloy and hydrogen gas used for simulation [18,29,49].

Parameter	MH alloy	H ₂ gas
Molecular mass, M (g.mol ⁻¹)	432	2.016
Thermal conductivity, k (W.m ⁻¹ K ⁻¹)	1.31	0.13
Rate constant, C_R (s ⁻¹)	59.187 (absorption) 9.57 (desorption)	–
Specific heat of alloy, C_p (J.kg ⁻¹ K ⁻¹)	419	14,283
Reaction enthalpy, ΔH (J.mol ⁻¹)	–30100	–
Reaction entropy, ΔS (J.mol ⁻¹ K ⁻¹)	111.74	–
Activation hydriding/dehydriding energy, A_e (J.mol ⁻¹)	21,170 (absorption) 16,473 (desorption)	–
Powder density, $\rho_{M,o}$ (kg.m ⁻³)	8300	–
Maximum storage percent (wt.%)	1.37	–
Porosity, ϵ_p	0.5	–
Slope constant, α_s, α_e	0.038, 0	–
Hysteresis constant, γ_H	0.137	–

When selecting a model domain for numerical modelling, it's important to maintain model symmetry with the minimum computational effort. Hence, for the SMTR, TFMTR, LFMTR, and DFMTR models, one-fourth of the section is considered, as these models maintain symmetry and the computational effort is also lower. But, in the SFMTR model, due to the spiral nature of the fins, one-sixth of the section is considered, as it offers minimal computational efforts while maintaining the symmetry. The considered computational domain for the different models is represented in Fig. 2. Furthermore, to simplify the model analysis, the

following assumptions are considered [23,29,49,50]:

- Hydrogen acts like an ideal gas.
- The hydrogen exchange takes place only through the porous filter.
- The thermo-physical properties of alloy and hydrogen are assumed to constant by temperature variations.
- The porosity and permeability of the bed are considered to be constant.
- The MH bed is assumed to be free of radiation effects.

Governing equations

Based on the above mentioned assumptions, the numerical model is governed by the mass, momentum, and energy conservation equations, represented by Eqs. (1)–(14).

Mass conservation. The mass balance equations of hydrogen gas and solid MH bed are expressed as,

$$H_2 \text{ gas : } \epsilon_p \frac{\partial \rho_f}{\partial t} + \nabla \cdot (\rho_f \vec{u}_f) = -(1 - \epsilon_p) \dot{S}_{m,a/d} \quad (1)$$

$$MH \text{ bed : } (1 - \epsilon_p) \frac{\partial \rho_M}{\partial t} = (1 - \epsilon_p) \dot{S}_{m,a/d} \quad (2)$$

where ρ_f, ρ_M designates the gas and solid phase densities, ϵ_p denotes the porosity and $\vec{u}_f$ is the gas velocity.

The density of gas varies with hydrogen exchange inside the bed and follows the ideal gas relation. The mass source term $\dot{S}_m$ is evaluated during absorption or desorption are expressed as,

$$\dot{S}_{m,a} = C_{R,a} \exp\left(\frac{-A_{e,a}}{R_u T}\right) \ln\left(\frac{P_f}{P_{eq}}\right) (\rho_{M,ss} - \rho_s) \quad (3)$$

$$\dot{S}_{m,d} = C_{R,d} \exp\left(\frac{-A_{e,d}}{R_u T}\right) \left(\frac{P_f - P_{eq}}{P_{eq}}\right) (\rho_s - \rho_{M,o}) \quad (4)$$

where $C_{R,a}/C_{R,d}$ denotes the reaction rate constant, $A_{e,a}/A_{e,d}$ represents the activation energies, P_f/P_{eq} is the inlet gas pressure and equilibrium pressure, and $\rho_{M,ss}/\rho_{M,o}$ denotes the saturated and unsaturated MH

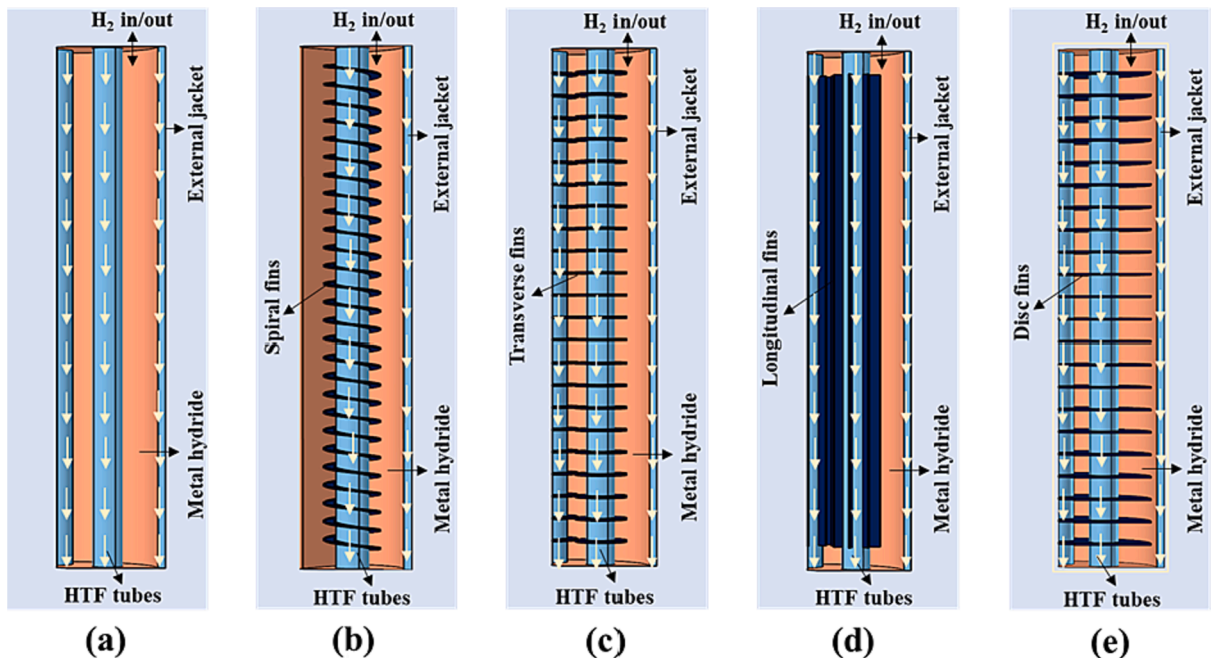


Fig. 2. The schematic of the computational domain (a) SMTR, (b) SFMTR, (c) TFMTR, (d) LFMTR, and (e) DFMTR models.

densities.

Momentum conservation. The velocity of gas within the porous domain is evaluated using the Darcy's law,

$$\vec{u}_f = \frac{-k_p}{\mu_f} \nabla P \quad (5)$$

where k_p shows the permeability of the porous domain and μ_f denotes the dynamic viscosity of hydrogen gas which is considered to be independent of temperature variations.

Energy conservation. The energy conservation equation assumes thermal equilibrium between the gas and the porous domain at each location and is used to evaluate the temperature within the MH bed, represented in Eq. (6).

$$(\rho C_p)_e \frac{\partial T}{\partial t} + (\rho C_p)_f (\vec{u}_f \cdot \nabla T) = k_e \nabla^2 T - Q_S \quad (6)$$

The mass source term ($\dot{S}_m$) value is positive and negative during the absorption and desorption processes, respectively. Also, reaction enthalpy term (ΔH) is considered negative for both the cases, as mentioned in Table 3. Therefore, the heat source term (Q_S) in Eq. (6) is provided with a '-' sign to make the heat source value positive in the case of the absorption process and negative in the case of the desorption process. $(\rho C_p)_e$ and k_e are the effective heat capacity and thermal conductivity inside the bed, represented by Eq. (7) and Eq. (8). Also, the heat source term (Q_S) is evaluated using Eq. (9).

$$(\rho C_p)_e = (1 - \varepsilon_p)(\rho C_p)_M + \varepsilon_p(\rho C_p)_f \quad (7)$$

$$k_e = (1 - \varepsilon_p)k_M + \varepsilon_p k_f \quad (8)$$

$$Q_S = (1 - \varepsilon_p) \dot{S}_{m,a/d} \left(\frac{\Delta H}{M_f} \right) \quad (9)$$

where M_f and ΔH represents the molecular mass and reaction enthalpy.

The energy balance between the extended surfaces (fins) and the reactor walls is expressed by Eq. (10).

$$(\rho C_p)_{rf} \frac{\partial T}{\partial t} = k_{rf} \nabla^2 T \quad (10)$$

where $(\rho C_p)_{rf}$ and k_{rf} represents the heat capacity and thermal conductivity of extended surfaces.

Equilibrium pressure. When the hydrogen interacts with the metal alloy at a certain pressure and temperature, the hydrogen molecules dissociate into hydrogen atoms and penetrate the surface of the alloy. Then, there exists a two phase region in which metal hydride formation occurs. The van't Hoff equation derived by Nishizaki et al., [51] mathematically defines the pressure-temperature relationship within the two phase region and the equilibrium pressure is evaluated using Eq. (11). However, the stresses developed within the metal hydride exhibit a hysteresis effect during hydrogen exchange processes. Further, the presence of inhomogeneity introduces sloping effects that must be taken into consideration. Therefore, by considering the hysteresis and sloping effects the equilibrium pressure is evaluated by Eq. (11).

$$\ln \left(\frac{P_{eq}}{P_a} \right) = \frac{\Delta S}{R_u} - \frac{\Delta H}{R_u T} + (\alpha_s \pm \alpha_e) \tan \left(\pi \left(\frac{x}{x_{max}} - 0.5 \right) \right) \pm \frac{\gamma_H}{2} \quad (11)$$

where '+' and '-' are considered for absorption and desorption half cycles, respectively. $\Delta H/\Delta S$ denotes the reaction enthalpy and entropy, α_s/α_e are the slope factors, γ_H is the hysteresis factor, and x_{max} is the maximum concentration capacity of hydrogen storage. The term 'x' denotes the concentration factor, and can be expressed by Eq. (12).

$$x = \frac{2(\rho_s - \rho_{M,o})M_M}{\rho_{M,o}N_M M_f} \quad (12)$$

where ρ_s denotes the MH density at time any 't', M_M is the molecular mass of alloy, N_M represents the number of hydrogen atoms, and $\rho_{M,o}$ is the unsaturated MH density.

Convective heat flux.. To reduce the computational time and efforts the present study uses the empirical correlations. The coefficient of heat transfer is evaluated by empirical correlations for the required flow rate, given in Eq. (13) [52].

$$\begin{aligned} \overline{Nu} &= 1.86 \left(Re_d Pr \frac{d}{l} \right)^{\frac{1}{3}} (Re_d < 2300) \\ \overline{Nu} &= 0.86 \left(\frac{d}{d_o} \right)^{0.16} \times 0.116 \left(1 + \left(\frac{d_h}{L} \right)^{\frac{2}{3}} \right) (Re_d^{\frac{2}{3}} - 125) Pr^{\frac{1}{3}} (2300 < Re_d < 10000) \\ \overline{Nu} &= 0.023 Re_d^{0.8} Pr^n (Re_d > 10000); n = 0.3 \text{ and } 0.4 \text{ for cooling and heating} \end{aligned} \quad (13)$$

$$\overline{Nu} = \frac{\bar{h} d_h}{k}$$

where $\overline{Nu}$, Re_d , Pr represents the Nusselt, Reynolds and Prandtl numbers, d , d_o and l is the inner diameter, outer diameter and length of the reactor, d_h is the characteristic length.

Model initial and boundary conditions

The bed temperature and gas variations are computed by considering the given initial and boundary conditions, represented in Table 4. A step function is employed to facilitate the easy transition of gas from the initial value of bed pressure (equilibrium) to the inlet or outlet pressure, given by Eq. (14).

$$P_{s,d} = P_{eq(a/d)} + (F(t))^* (P_{a/d} - P_{eq(a/d)}) \quad (14)$$

Table 4
Initial and Boundary conditions.

Initial Conditions		
Parameter	Absorption Value	Desorption Value
$T(r, \theta, z)$	$T_{ini,a}$	$T_{ini,d}$
$P(r, \theta, z)$	$P_{eqi,a}$	$P_{eqi,d}$
$\rho_s(r, \theta, z)$	$\rho_{Mi,a}$	$\rho_{Mi,d}$
$x(r, \theta, z)$	$x_{i,a} = 0.05 x_{max}$	$x_{i,d} = 0.95 x_{max}$
Boundary Conditions		
Type	Condition	
Outer walls are considered adiabatic	$\frac{\partial T}{\partial n} = 0$	
Non-permeability and no slip on the walls of HTF tubes	$\vec{u}_r = \vec{u}_z = 0$	
Hydrogen entry and flow restrict condition	At inlet of reactor, $P = P_{f,d}$ and on walls of MH reactor, $\frac{\partial p}{\partial n} = 0$ and $\vec{n} \cdot \rho \vec{u} = 0$	
Convective heat flux condition	At the interface of tube/shell and the reactor: $(-k \left[\frac{\partial T}{\partial n} \right])_{Alloy} = (-k \left[\frac{\partial T}{\partial n} \right])_{tube/shell}$ At the surface of HTF tube/shell: heat flux condition $-\vec{n} \cdot (-k \left[\frac{\partial T}{\partial n} \right])_{tube/shell} = \bar{h}(T_{HTF} - T)$	

where function $F(t)$ varies with time at a step of $t = 0.05$ s.

Mesh and time step independence

The accuracy of the results obtained from the numerical model significantly depends on the selected grid size and time step value used for simulation. Hence, it is vital to carefully choose the appropriate mesh elements and time step values for accurate results. The mesh independence test (MIT) was performed on the DFMTR model to evaluate the impact of varying the mesh elements during the absorption process.

Three different mesh sizes, namely normal (1,225,137 elements), fine (1,983,338 elements), and finer (2,615,873 elements), were considered for the test. Fig. 3a shows the average bed temperature and concentration factor with different mesh sizes. The results indicate that

the outcomes for all mesh sizes are nearly identical and overlapping. Hence, for all simulations, fine mesh is taken into consideration. Fig. 3b shows that the outcomes of average bed temperature and concentration factor for different time steps are nearly identical, indicating that the simulations are independent of the time step. Therefore, the adaptive time stepping scheme is adopted for all simulations, as it drastically reduces the simulation time without affecting the accuracy of outcomes. To solve the time step, the backward differentiation formula (BDF) method is used with an initial step of 0.001 s. Further, the relative tolerance value is set to 10^{-5} for all simulations. The highly nonlinear heat and mass transfer problem is solved using a time-dependent solver, PARADISO [53].

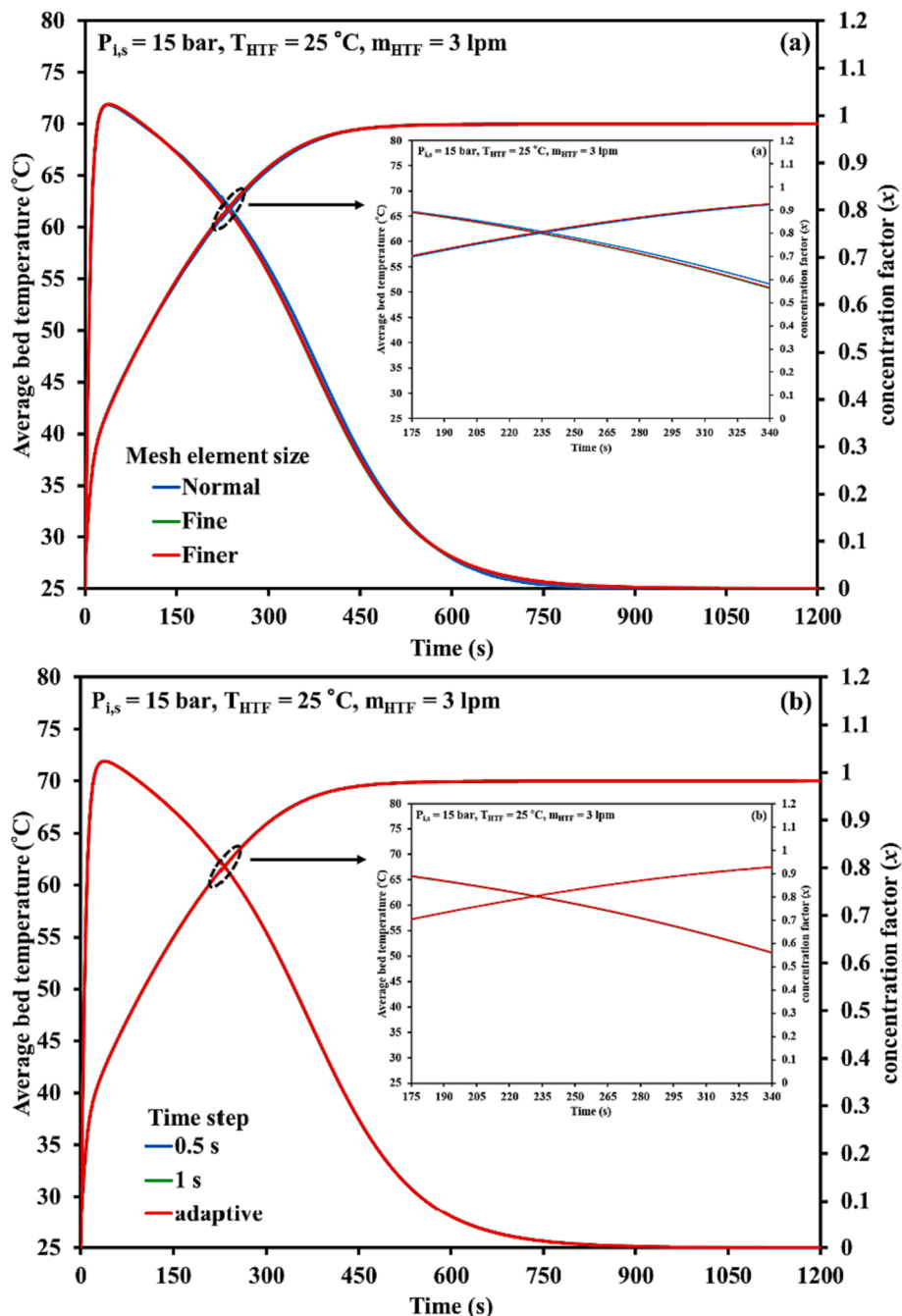


Fig. 3. Impact of (a) Mesh size and (b) time step size on the average bed temperature and concentration factor.

Model validation

The originality of the simulated model can be evaluated by comparing the numerical outcomes with actual experimental results available in the literature. The findings of this numerical study under similar operating conditions are compared with the experimental work of Muthukumar et al. [54] and Singh et al. [18]. Fig. 4 demonstrates the comparison results for the quantity of hydrogen absorbed and the temperature of the MH bed. It was observed that the numerical model precisely agreed with the outcomes of the experimental data. However, the maximum error values between experimental and numerical outcomes in calculating the relative quantity of hydrogen absorbed and bed temperature were estimated as $\pm 0.7\%$ and ± 1.9 K, respectively. The possible reason for this could be that the numerical model assumes

adiabatic conditions on the walls (no losses to the atmosphere), whereas in actual cases, such losses are inevitable.

Results and discussion

This section presents a detailed analysis of the proposed reactor design for large capacity hydrogen storage. In Section 3.1, five reactor designs are analysed during the absorption and desorption half cycles. The outcomes of these reactor designs are compared to determine the most efficient reactor design. Further, Section 3.2 deals with the design optimization of the proposed reactor to understand the effect of the fin distribution on the reactor's performance. Furthermore, a detailed analysis of the effect of the operating parameters on the performance of the proposed reactor is discussed in Section 3.3. Moreover, Section 3.4

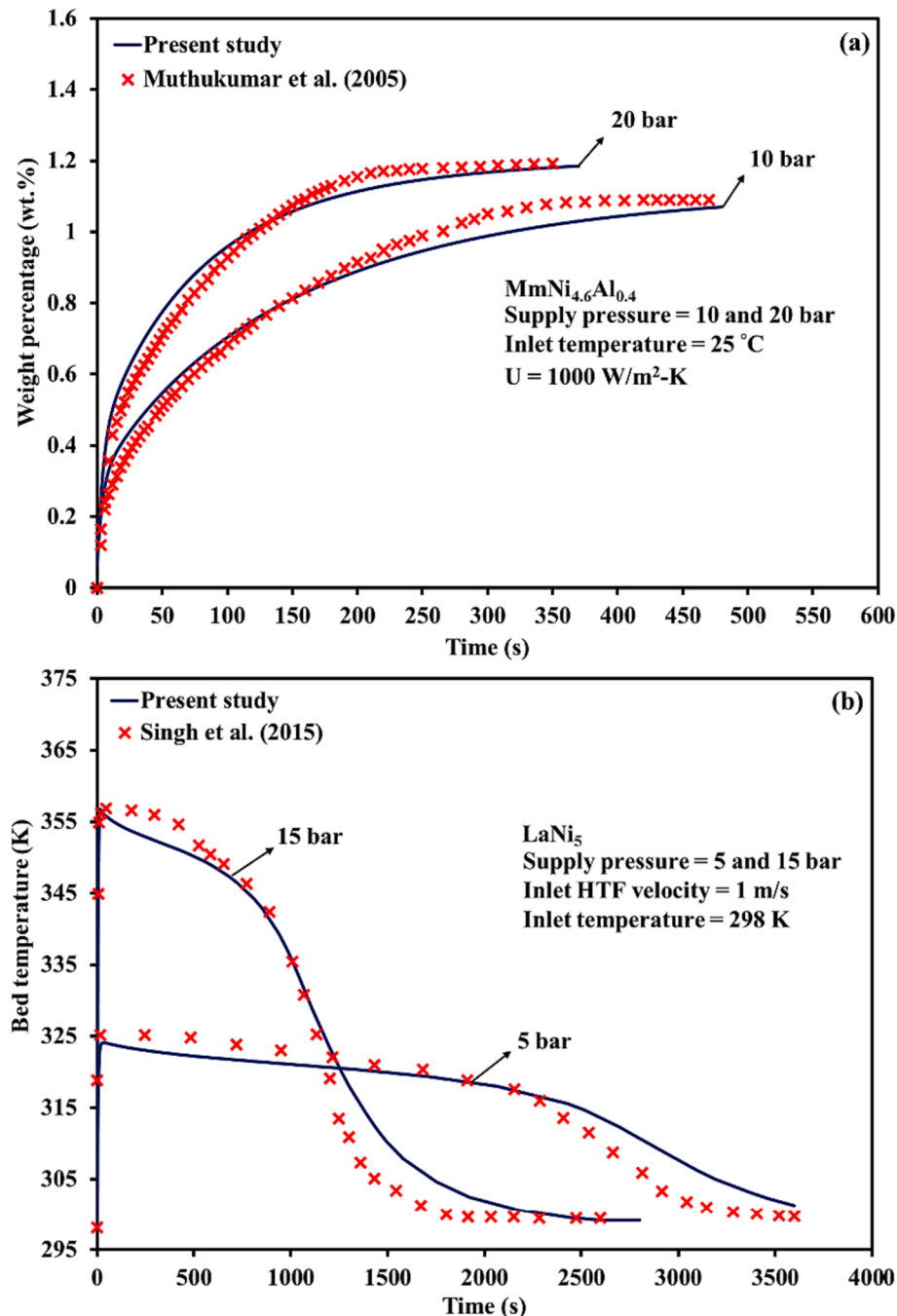


Fig. 4. Validation of simulation outcomes with experimental data (a) Muthukumar et al., (2005) and (b) Singh et al., (2015).

deals with the scale-up methodology and the comparative study of the proposed reactor design with various studies reported in the literature.

Reactors comparison study

The traditional and widely adopted reactor structure for hydrogen storage is the shell and tube type. It is obvious that adding more tubes within the reactor increases the internal surface area and, thus, its hydrogen storage performance. However, for a given reactor capacity, this also results in greater bulkiness, complexity, and a decrease in gravimetric storage density. The use of extended surfaces is a substitute strategy that improves performance without significantly increasing reactor mass. However, the type and distribution of extended surfaces

are important concerns to achieve the most favorable reactor conditions. Therefore, it is crucial to carry out a comparative study on various fin structures to achieve the excellent reactor performance. Hence, the performance of five reactor models attaining the same fin volume, namely, straight multi tube reactor (SMTR), transverse finned multi tube reactor (TFMTR), spiral finned multi tube reactor (SFMTR), longitudinal finned multi tube reactor (LFMTR), and disc finned multi tube reactor (DFMTR) is investigated and compared during the absorption and desorption half cycles. The SMTR is regarded as the baseline model, and the performance of other models is assessed in comparison to it. Moreover, the mass ratio (ratio of alloy mass to the reactor mass) is somewhat lower by 5.7% when expanded surfaces are taken into account compared to the baseline reactor. For absorption and desorption half cycles, the

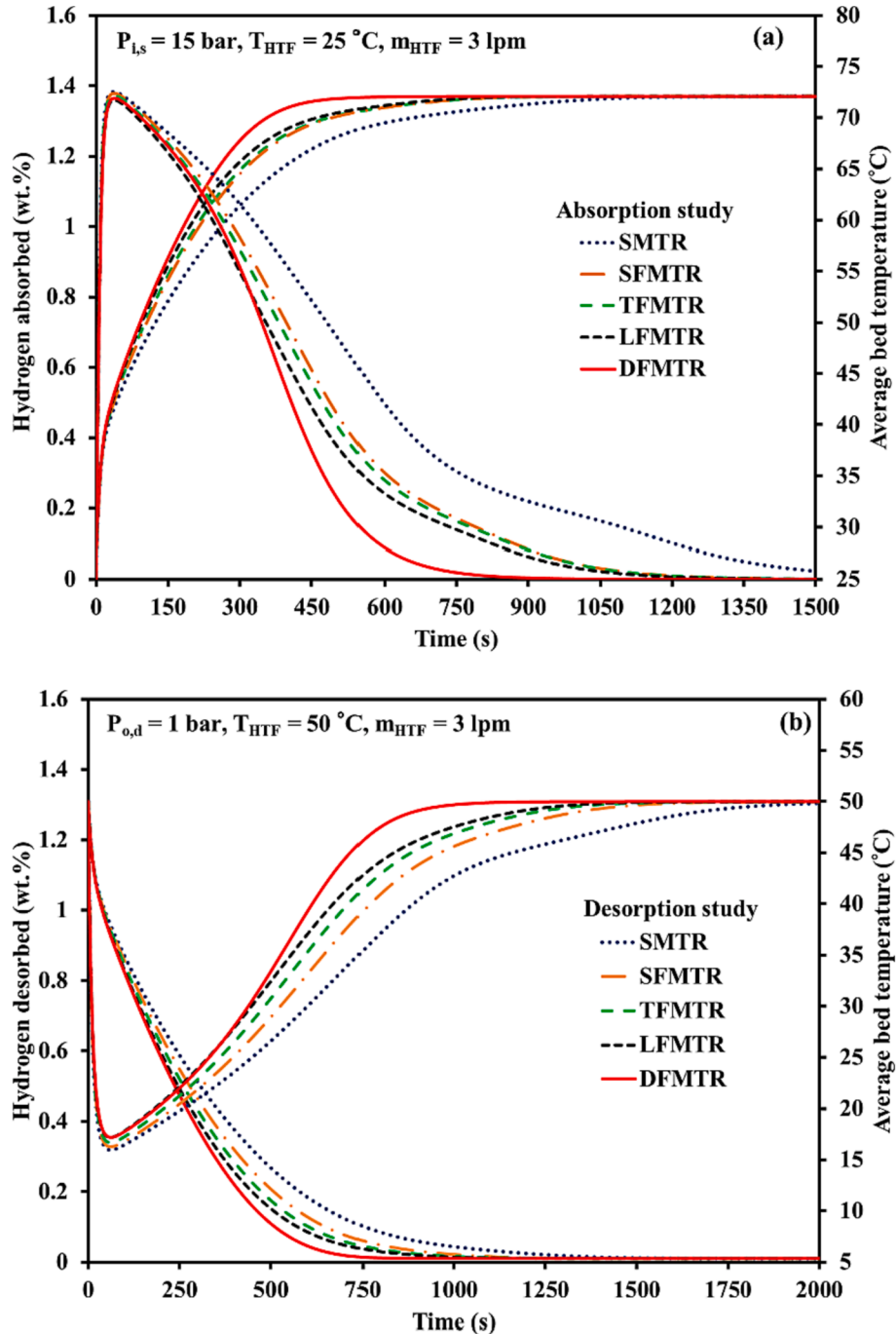


Fig. 5. Comparison between reactor models during (a) absorption half cycle, (b) desorption half cycle and (c) percentage reduction in time of different models to absorb or desorb 90% of hydrogen compared to model 1.

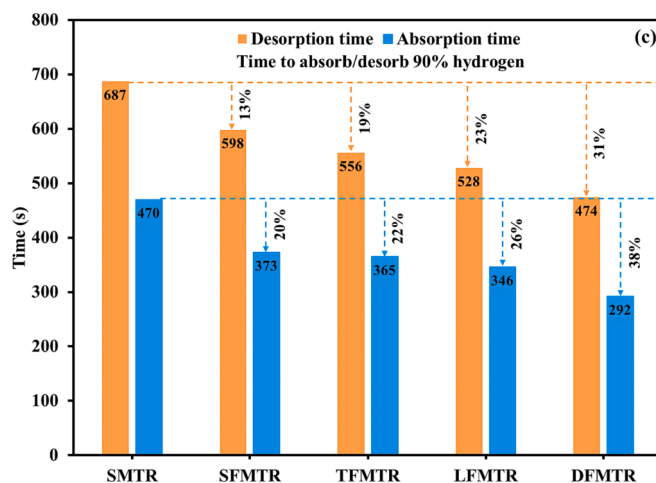


Fig. 5. (continued).

value of the heat transfer coefficient is estimated corresponding to the flow rate of 3 lpm, and its value is taken as the same for all models due to similar internal geometry. The estimated values of heat transfer coefficients for the tube and external jacket are $457 \text{ W.m}^{-2}\text{K}^{-1}$ and $372 \text{ W.m}^{-2}\text{K}^{-1}$, as per Eq. (13).

During the absorption half cycle, the inlet hydrogen gas pressure and cooling fluid temperature are set at 15 bar and 25°C , respectively, for all the simulated models. Initially, there is a significant pressure difference, which is the difference between the gas pressure and the equilibrium pressure of the reactor, which results in rapid hydrogen absorption by the metallic alloy. However, the heat is produced inside the reactor as a result of hydrogen being absorbed by the metallic alloy. This heat causes an immediate rise in the reactor's temperature, which exponentially increases the reactor's equilibrium pressure, as per Eq. (11). As a result, the pressure difference starts dropping, which slows down the rate of absorption. Finally, the pressure difference becomes zero, and the hydrogen absorption half cycle is completed. The average reactor temperature and hydrogen absorbed (wt.%) for all models are represented in Fig. 5a. It is observed that all models show similar performance initially and absorb a weight percentage (wt.%) of 0.32 in the first 10 s. Later, it is observed that the DFMTR model absorbs hydrogen at a faster rate than other models. The DFMTR model absorbs the maximum storage capacity, i.e., 1.37 wt.% of hydrogen in 795 s, which is 46%, 30%, 28%, and 27% faster than the SMTR, SFMTR, TFMTR, and LFMTR models. Also, the maximum temperature of 72°C is reached by all the models in just 32 s after absorption starts, as shown in Fig. 5a. It is seen that the drop in the bed temperature of the DFMTR model is very fast due to improved heat transfer, and the bed took 995 s to return to the initial temperature of the HTF, whereas the LFMTR model, TFMTR model, SFMTR model, and SMTR model take 1404 s, 1438 s, 1452 s, and 1928 s, respectively. However, the heat transfer rate at the end of the absorption or desorption reactions was extremely slow; therefore, the absorption process time was considered to be the time required to reach the initial temperature within a temperature tolerance of 0.2°C , i.e. 25.2°C . The time required to reach the cooling fluid temperature by DFMTR is reduced by 1.93 times as compared to SMTR.

The desorption half cycle encompasses the hydrogen discharge process. In the present study, the hydrogen is discharged to the atmosphere (1 bar pressure) at a heating fluid temperature of 50°C . When the process initiates, the discharge rate is very high due to the higher pressure difference between the reactor's equilibrium and atmosphere pressures. Also, there is a sudden drop in reactor temperature due to the endothermic nature of the process, which reduces the pressure difference; therefore, a reduction in the discharge rate is observed. Later, the reactor temperature starts increasing by taking heat from the heating fluid and finally reaches the initial conditions. Fig. 5b shows the time

behavior of average reactor temperature and hydrogen desorbed (wt.%) for all models. Initially, all models behaved similarly and discharged 0.15 wt.% of hydrogen in the first 10 s.

However, in a later stage, the DFMTR model was found to be superior and discharges all the stored hydrogen in 1060 s, whereas the SMTR, SFMTR, TFMTR, and LFMTR models take 2245 s, 1622 s, 1494 s, and 1439 s, respectively. The lowest temperature attained by all models is 16.5°C in the initial 60 s. The DFMTR model takes 1317 s to reach the heating fluid temperature, which is 48%, 33%, 27%, and 24% less than the SMTR, SFMTR, TFMTR, and LFMTR models. Similar to the absorption process, the desorption process time was considered to be the time required to reach the initial temperature within a temperature tolerance of 0.2°C , i.e. 49.8°C . The percentage reduction in time of the absorption and desorption half cycles to absorb or desorb 90% of hydrogen for all models compared to the baseline model is represented in Fig. 5c. It is concluded from the above discussion that the large capacity DFMTR model is efficient and saves 38% and 31% of the time for absorbing and desorbing 90% of hydrogen compared to the baseline model. However, the geometrical parameters inside the reactor should be in an optimized position to achieve excellent performance. Therefore, it would be worthwhile to do more research to observe the distribution of extended surfaces inside the reactor.

The different cross-sectional cuts along the axial direction represent the temperature distribution within the reactors, as shown in Fig. 6. The cooler regions (blue regions) and hotter regions (red regions) signify the low and high temperature regions inside the MH reactors. The pressure difference is higher at lower bed temperatures during hydrogen storage and higher at high temperatures during hydrogen release. Therefore, the cooler regions (blue regions) and hotter regions (red regions) in Fig. 6 show excellent performance during the absorption and desorption processes, respectively. From Fig. 6, it can be observed that the deviations occur only in the radial direction for each reactor, while there is a uniform temperature distribution along the axial direction. During the absorption process, it is observed that in the case of SMTR, hotter regions are more visible, whereas in the case of DFMTR, cooler regions are more visible compared to the other reactors due to the excellent heat transfer provided by the disc fins. Therefore, DFMTR demonstrates superior performance among all the reactors during the absorption process. Similarly, during the desorption process, cooler regions are more visible in SMTR, while regions closer to hotter regions are observed in DFMTR, as shown in Fig. 6b. Therefore, the predicted performance of DFMTR is found to be superior during the desorption process as well.

Design optimization study

In this section, the design optimization methodology of the proposed

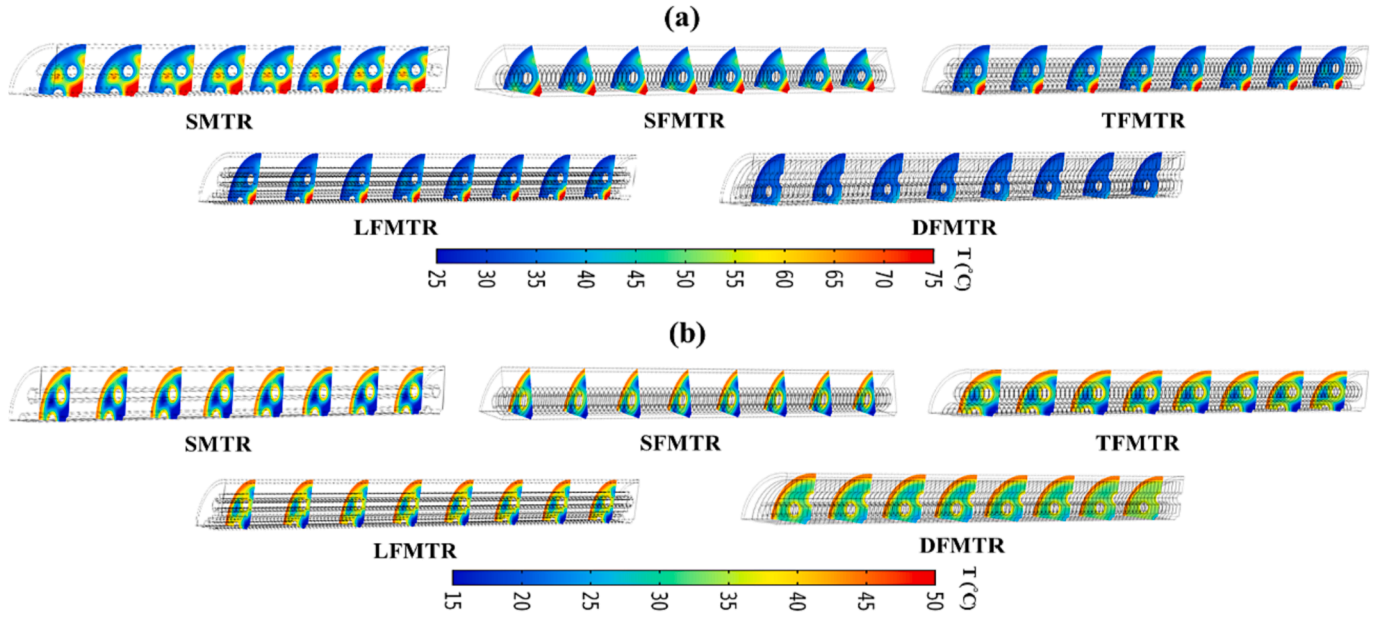


Fig. 6. Temperature distributions of various reactors at 500 s during: (a) absorption process and (b) desorption process.

DFMTR model has been discussed in detail. Further, the impact of fin distribution, thickness of fin, and external jacket on performance is investigated. The present work adopts a design approach that efficiently optimizes the heat transfer performance of the reactor, taking into account essential factors such as size, weight, strength, and cost. The flow chart for the design strategy of the present study reactor is represented in Fig. 7. Firstly, the objective of reactor design based on the particular application is fixed. After this, the selection of type and quantity of alloy required based on their storage capacity (wt.%) is fixed. Thereafter, the diameter and number of tubes are judiciously selected based on the established tube selection criteria. Based on the quantity of alloy and required strength, the reactor dimensions (diameter and length of

reactor) are finalized. To further enhance the performance of the reactor, the appropriate fin type is selected. Thereafter, to achieve the best performance, the optimization of fin distribution based on pitch and thickness is done. At last, the optimum operating condition based on appropriate gas pressure and fluid temperature is selected, and the design is finalized. It is to be noted that the present design methodology is independent of the type of fluid, the type of alloy, and its hydrogen storage capacity.

Effect of axial pitch (a_p)

The axial pitch represents the uniform distribution of the number of fins along the length of the reactor. Heat transfer within the reactor

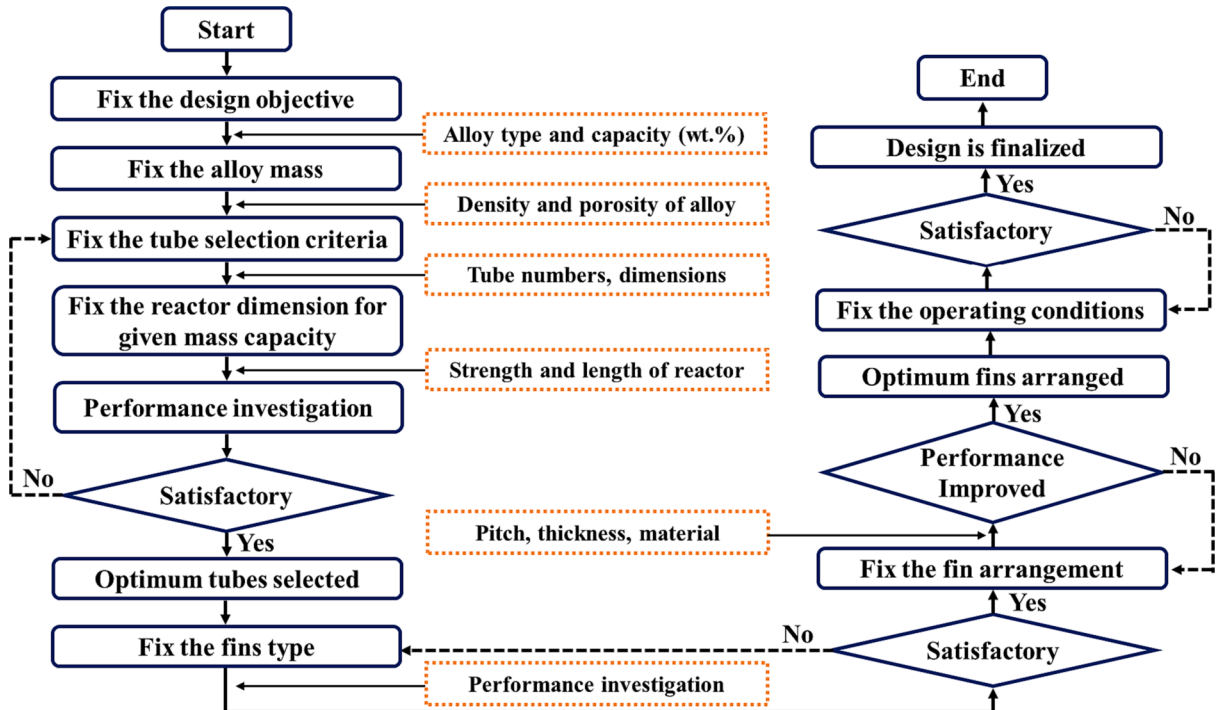


Fig. 7. Flowchart showing design methodology for present study reactor.

primarily relies on the number of fins and tubes. Notably, augmenting the number of fins serves to diminish thermal resistance by increasing the heat transfer area, thereby enhancing the reactor's thermal performance. However, it's important to note that an excessive increase in the fin numbers not only diminishes the gravimetric capacity but also introduces greater complexity and bulkiness into the system. Hence, it is crucial to carefully choose the number of fins to enhance performance without significantly compromising storage capacity. The effects of varying the axial pitch on the reactor performance under the same operating condition are represented in Fig. 8.

The axial pitch varied from 9 mm to 15 mm in the step of 1.5 mm, while the thickness of the fin is fixed at 0.5 mm. It is observed that reducing the axial pitch from 15 to 9 mm reduced the time needed to attain 90% of the storage and release capacities by 15% and 11%,

respectively. Additionally, during the absorption and desorption half cycles, there is a 10% and 12% reduction in the amount of time needed to achieve bed temperature at its initial conditions. Beyond 10.5 mm axial pitch, it is seen that the curves are almost overlapping. As a result, 10.5 mm is the optimum axial pitch value, which comprises 94 disc fins that are evenly spaced apart.

Effect of fin thickness (t_f)

The effect of fin thickness on reactor performance during the absorption and desorption half cycles is presented in Fig. 9. The relatively small distance between the curves on the absorption and desorption plots indicates that the performance of the reactor is not greatly enhanced by increasing the fin thickness.

When the thickness is raised from 0.25 to 1.25 mm, the time saved

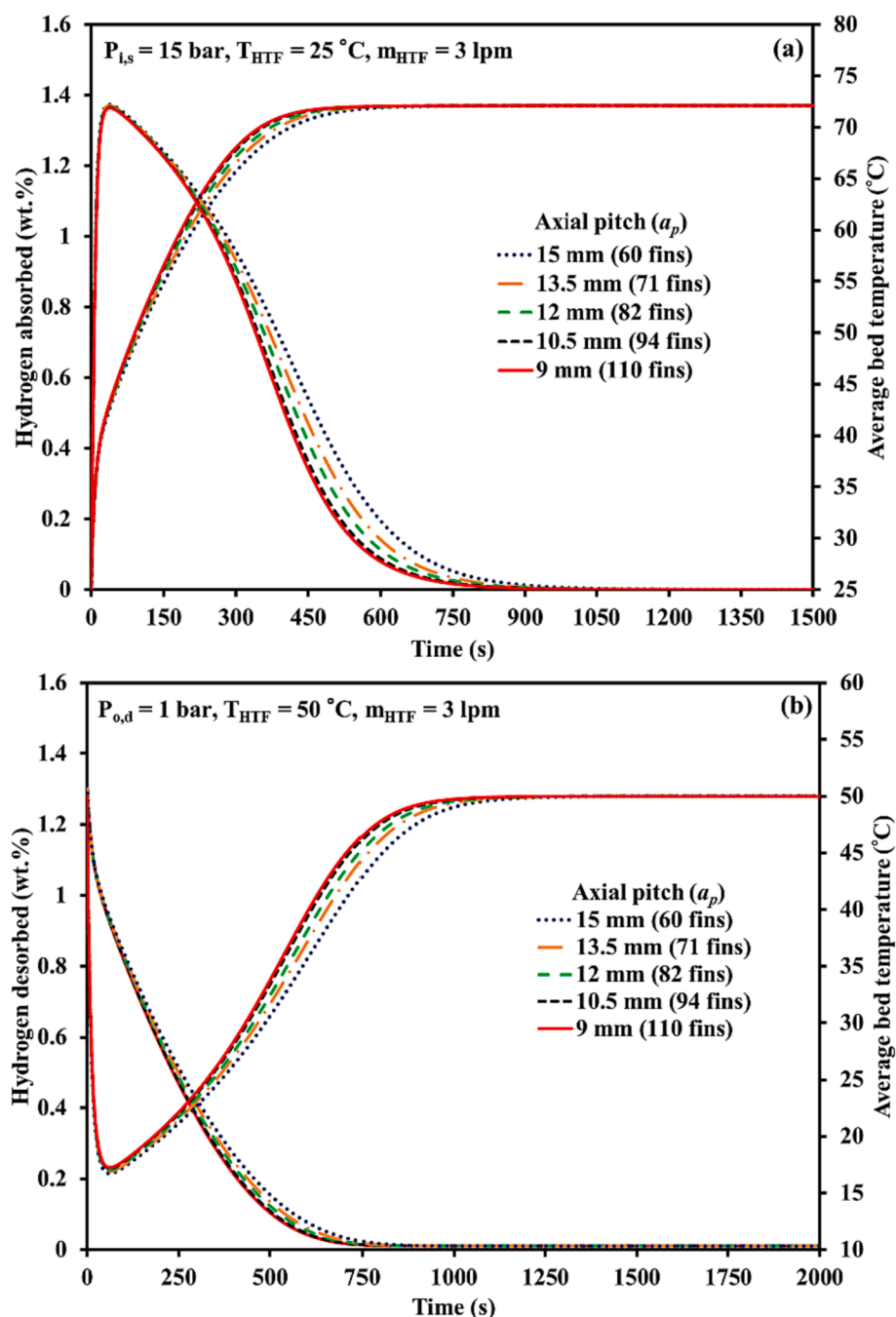


Fig. 8. Impact of axial pitch on reactor performance during: (a) absorption and (b) desorption half cycle.

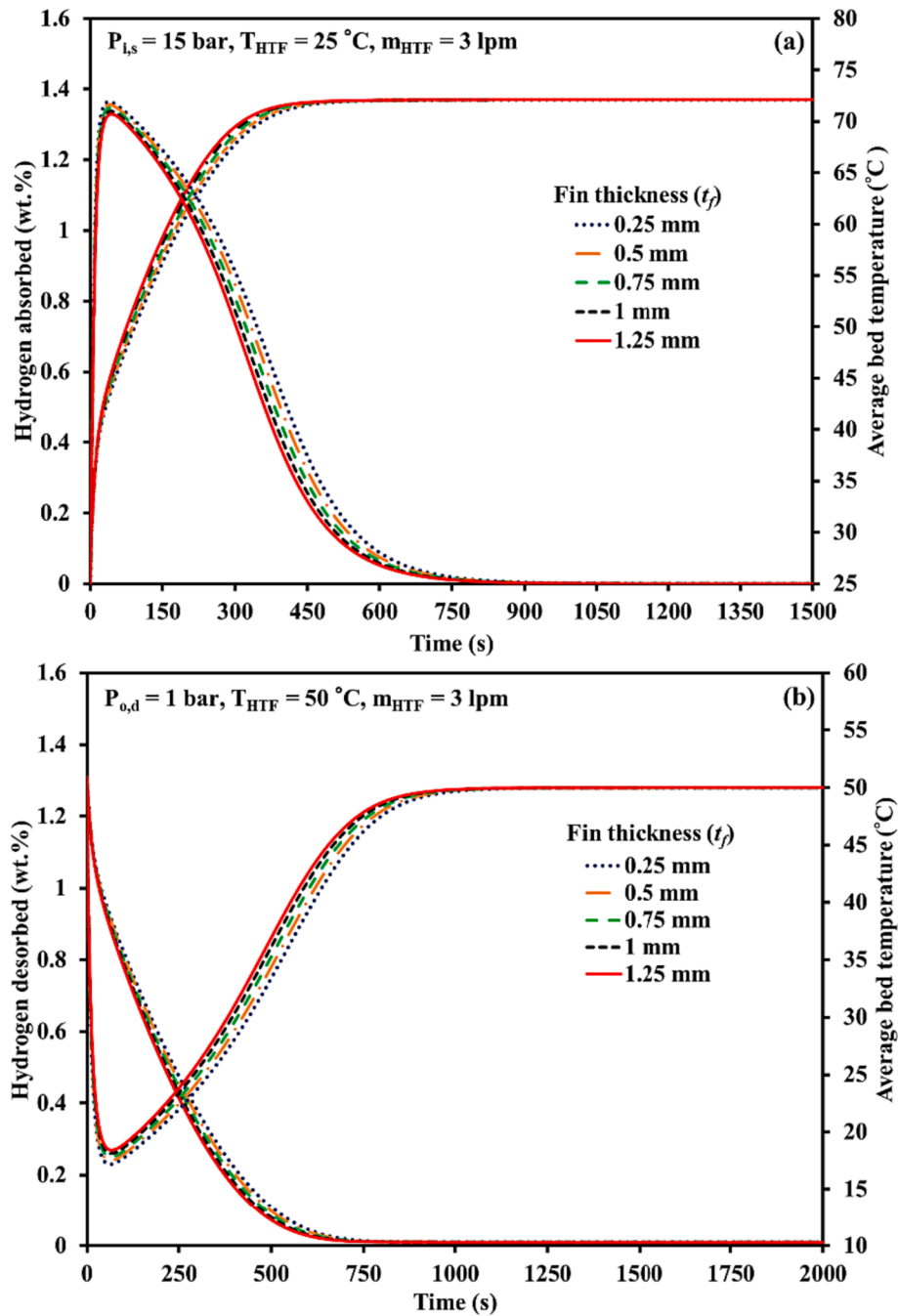


Fig. 9. Impact of fin thickness on reactor performance during: (a) absorption and (b) desorption half cycle.

for storing and releasing 90% of hydrogen is 7% and 6%, respectively, during the absorption and desorption half cycles, shown in Fig. 9a. Further, during the absorption and desorption half cycles, there is a 2% and 3% reduction in the amount of time needed to achieve bed temperature at its initial conditions, as illustrated in Fig. 9b. However, the mass of fins is also increased five times, which is not justified compared to the enhancement achieved. Therefore, considering the standard availability and performance, the optimum fin thickness value is considered to be 0.5 mm.

Effect of an external jacket

The performance of DFMTR is analyzed throughout the absorption and desorption half cycles to visualize the impact of the external jacket on the reactor's performance. The use of an exterior jacket not only reduces reaction blind spots around the reactor's end corners by

enhancing heat transfer area but also effectively cools or heats the reactor, which improves the reactor's performance drastically. Fig. 10 depicts the curves for the bed temperature and hydrogen absorbed or desorbed through the absorption and desorption half cycles. The time required to store and release 90% hydrogen with an external jacket is 380 s and 610 s, which was significantly reduced by 30% and 28% compared to the case without an external jacket during the absorption and desorption half cycles, respectively. However, the time savings to reach bed temperature at its initial conditions are 38% and 44% with the external jacket case during the absorption and desorption half cycles. Therefore, it is recommended to use an external jacket to achieve excellent reactor performance.

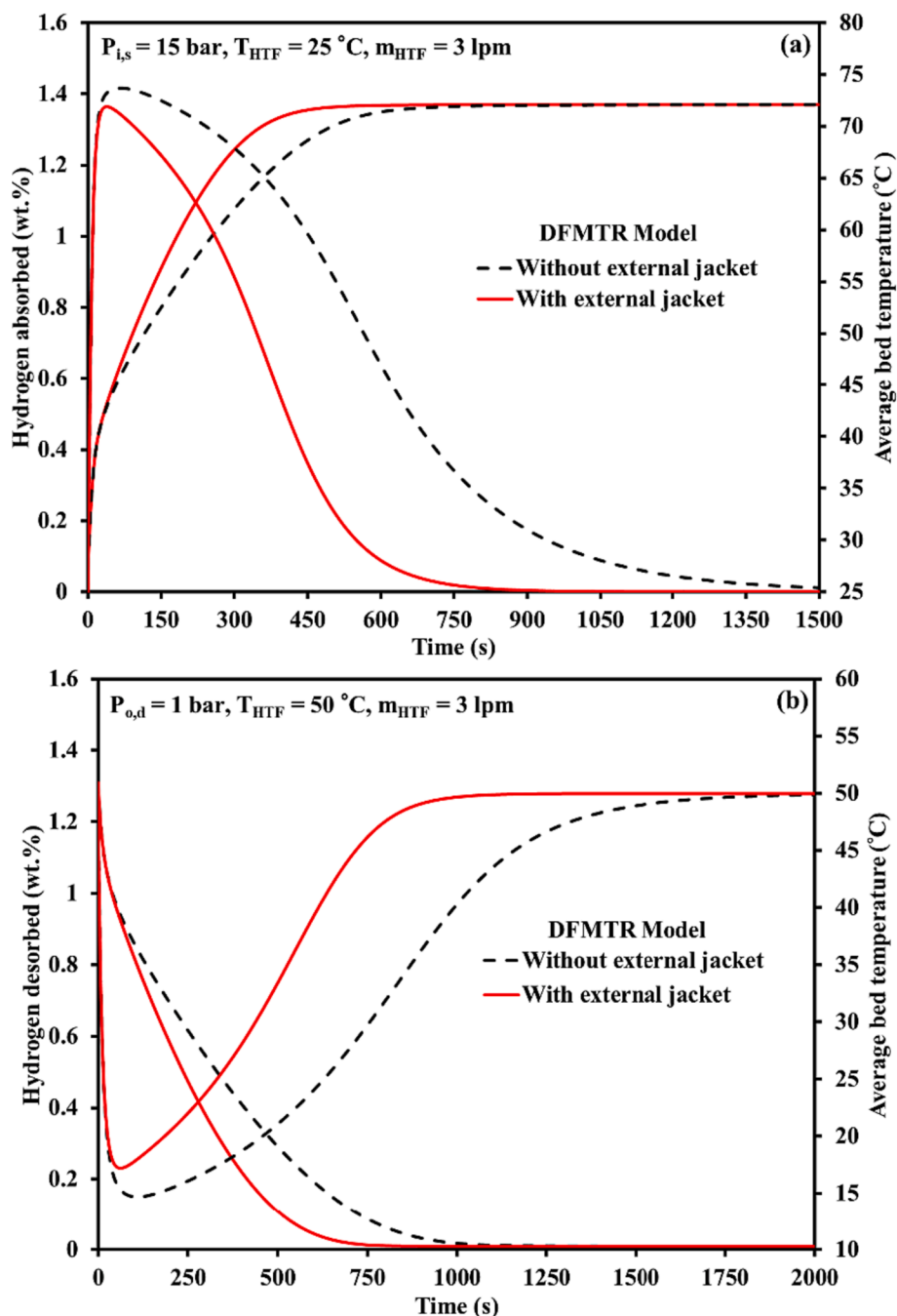


Fig. 10. Impact of an external jacket on DFMTR model performance during: (a) absorption and (b) desorption half cycle.

Operating parametric assessment

The reactor's performance is significantly influenced by two key parameters: the inlet or outlet pressure and the temperature of the cooling or heating fluid. Therefore, this section presents the effect of these parameters on the performance of the proposed optimized reactor (DFMTR model) during the absorption or desorption half cycles in detail.

The inlet gas pressure greatly affects the pressure difference (the difference between the inlet supply gas pressure and the reactor's equilibrium pressure); therefore, its effect on the absorption performance of the DFMTR model is studied. This study varied the inlet gas pressure from 10 bar to 25 bar while maintaining the cooling fluid temperature and flow rate at 25 °C and 3 lpm. Fig. 11a shows the

influence of inlet gas pressure on the average bed temperature and the amount of hydrogen absorbed. The observation revealed that an increase in gas pressure increases the pressure difference, which results in a faster absorption rate. However, this higher pressure also leads to increased heat generation within the system, consequently causing a rise in reactor temperature. The absorption time to reach 90% of absorption capacity decreased by 53% when the gas pressure was increased from 10 bar to 25 bar. However, the reduction becomes only 9%, which was considered insignificant, when the gas pressure exceeds 20 bar. Further, a 24% reduction was observed in time to reach the initial cooling temperature when pressure increased from 10 to 20 bar. Therefore, the optimized gas pressure value during the absorption process is chosen to be 20 bar.

Further, the impact of cooling fluid temperatures on the hydrogen

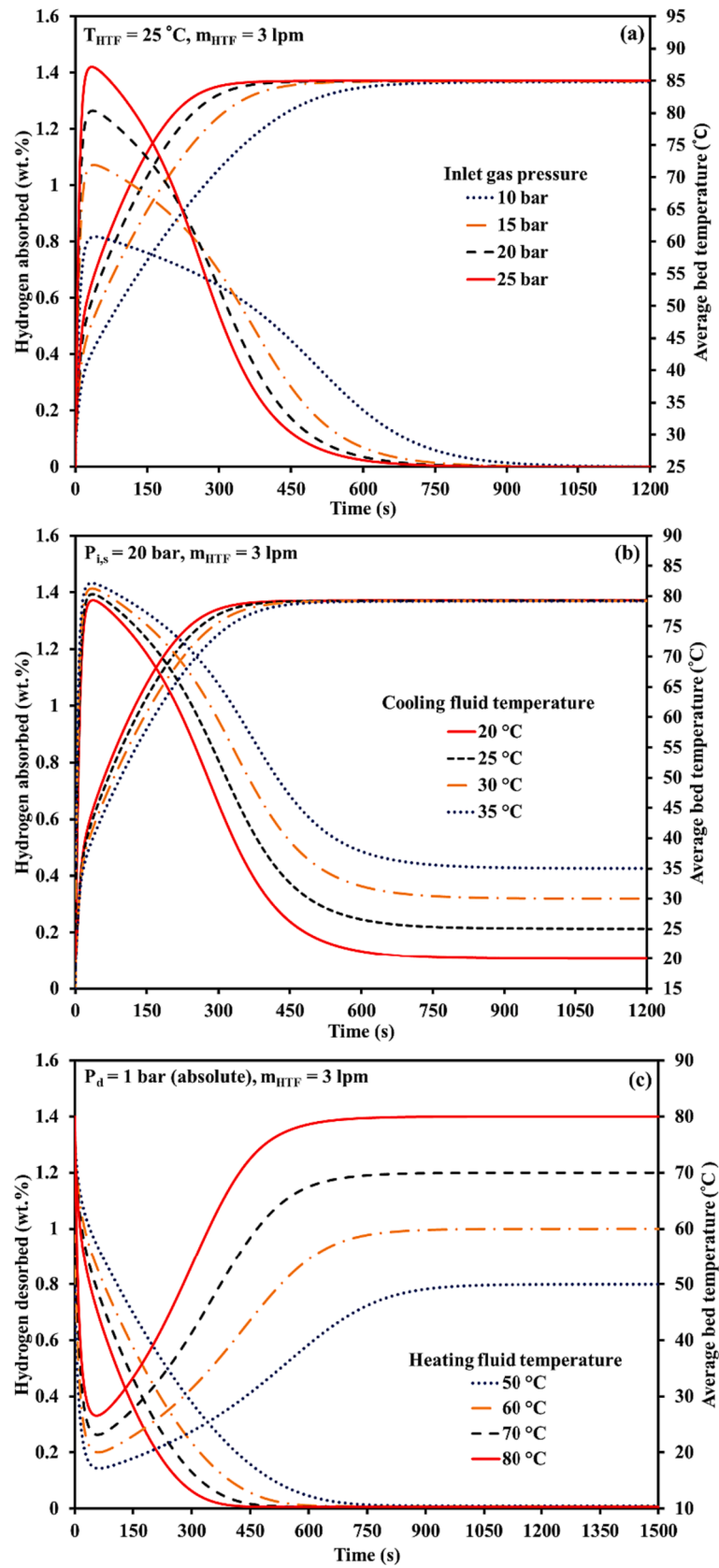


Fig. 11. (a) Impact of the inlet gas pressure on absorption process, (b) Impact of the cooling fluid temperature on absorption process and (c) Impact of the heating fluid temperature on desorption process.

storage performance of the reactor is investigated at an optimized gas pressure of 20 bar. As per van't Hoff equation (Eq. (11)), decreasing the temperature decreases the equilibrium pressure within the MH bed; therefore, the pressure difference (the difference between the inlet supply gas pressure and the reactor's equilibrium pressure) rises, which increases the reaction rate of the process. Therefore, the outcomes shown in Fig. 11b reveal that at a low cooling fluid temperature, the reaction rate enhances and the hydrogen is absorbed by the reactor in a faster way. Also, the bed temperature of the MH bed reaches the initial conditions quickly at lower temperatures.

The result shows a saving of 16% in absorption time for reaching 90% capacity when the temperature is reduced from 35 °C to 25 °C. Further, there is a 6 % saving in time to reach 90% capacity when operated below 25 °C. However, the reactor generally operates at ambient temperature (i.e., 25 ± 1 °C) which is considered to be the optimum value because the improvement is insignificant in comparison to the cost involved in cooling and maintaining temperatures below the atmospheric level.

Furthermore, the influence of heating fluid temperature on the reactor's release performance is analyzed. Desorption at higher temperatures leads to an increase in the equilibrium pressure within the bed, as per van't Hoff Eq. (11). Due to this, the pressure difference (the difference between the equilibrium pressure of the bed and the discharge (ambient) pressure) increases, resulting in a faster reaction rate at higher temperatures. The impact of heating fluid temperature on the bed temperature and amount of hydrogen desorbed is represented in Fig. 11c. It is observed that the reactor released 90% hydrogen in 478 s when the fluid temperature is 50 °C. However, the time for the same is dropped by 22%, 36%, and 48% when the desorption is done at fluid temperatures of 60 °C, 70 °C, and 80 °C, respectively. Further, the reactor took 1317 s, 1160 s, 1113 s, and 1056 s to reach the bed temperature at their initial conditions of 50 °C, 60 °C, 70 °C, and 80 °C, respectively. When fluid temperature is raised from 70 °C to 80 °C, it is seen that the curves get closer and time is saved by only 5%, which is not enough when compared to the expense involved in heating the reactor. Therefore, the optimum heating fluid temperature for the proposed reactor is considered to be 70 °C.

Scale-up methodology and comparative study with literature

The scale-up methodology primarily relies on the desired hydrogen storage capacity. Rather than utilizing a single large reactor, the present study suggests a portable design approach in which multiple small reactors can serve a various purpose in a variety of applications. The hydrogen economy encompasses the generation, storage, and utilization of hydrogen, as illustrated in Fig. 12.

Numerous proposed disc finned multi tube reactors can couple in parallel to accommodate a substantial hydrogen inventory generated by the electrolyzers that can be utilized in various applications, depending on the required capacity of the storage system. This portable design

approach provides the ability to utilize a portion of the whole storage system according to demand and an easier way to repair or replace the reactor in case of any leakage, which enables operational and maintenance flexibility. Additionally, this strategy aids in lowering the dangers associated with inadvertent leaking and offers an effective way to operate the system at a lower capacity whenever needed. Therefore, the suggested portable design strategy offers an efficient way of storing hydrogen without affecting the system's performance. Further, a comparison is made between the predicted performance of the proposed DFMTR model and the recently reported reactors in the literature. The performance of the reactor during the absorption and desorption processes mainly depends on the design of the reactor. The straightforward and effective method of assessing the hydrogen storage reactor performance is to compare the 90% hydrogen storage time. Therefore, the present section compares the predicted performance of the DFMTR reactor with the reactors used in the literature during the absorption process. The time required to store 90% of hydrogen is compared with various reactors that contained different quantities of LaNi₅ alloy. The

Table 5
Recent reported reactor designs considered for the comparison.

Author, Year [Ref.]	Reactor design type	LaNi ₅ alloy quantity	Absorption condition (inlet pressure, HTF temperature and flow rate)	Time to store 90 % hydrogen
Present work	Disc finned multi tube reactor	19.4 kg	15 bar, 25 °C, and 3 lpm	292 s
Chandra et al. 2020 [29]	Multi tube tapered finned reactor consist of 6 cooling tubes	5 kg	15 bar, 20 °C, and 5 lpm	375 s
Jana et al. 2021 [42]	Tube bundle reactor (19 tubes): Single tube having the outer diameter, thickness and length of 25.4 mm, 1.2 mm and 1000 mm is considered for simulation	1.4 kg	Simulated under the same operating conditions as the present study.	585 s
Singh et al. 2017 [55]	Reactor consists of two U-shaped cooling tubes attached to 13 radial fins	1.2 kg	15 bar, 25 °C, and 2 lpm	610 s
Keshari and Maiya 2018 [56]	Pin fin type reactor consist of 16 cooling tubes	1.56 kg	15 bar, 25 °C, and 6.75 lpm	320 s

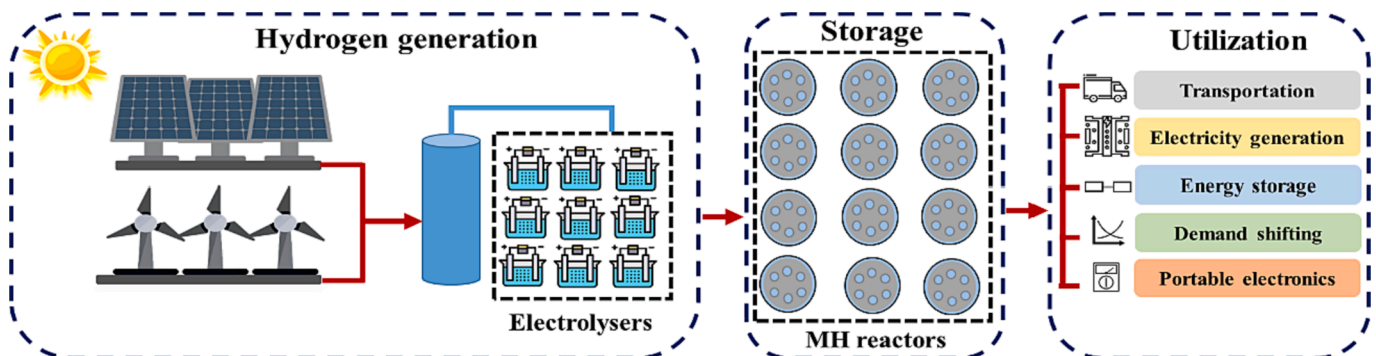


Fig. 12. Schematic of hydrogen economy.

specifics of the studies chosen for comparison, including their alloy quantity and operating conditions, are presented in Table 5.

The predicted performance of the proposed reactor (DFMTR model) is found to be superior to other reactors reported in the literature. The proportion of absorption time saved by the present study reactor compared to other designs is presented in Fig. 13. Due to the large surface area provided by the disc fins in the proposed reactor, a significant amount of reaction time was seen to be saved. The significant improvement can be attributed to the judicious optimization of the design and the utilization of extended surfaces, which greatly enhance internal heat transfer. The comparative analysis makes it evident that utilizing extended surfaces in the MH bed is a preferable choice over employing numerous tubes. Additionally, the inclusion of extended surfaces results in a lighter reactor and reduces complexity in comparison to the multiple tubes. Moreover, the author believes that the proposed reactor is efficient serve a variety of applications with a large hydrogen inventory.

Conclusions

In the present work, a novel disc finned multi-tube reactor (DFMTR) is proposed to enhance the reactor performance by considering the higher mass ratio (ratio of alloy mass to reactor mass). Firstly, the numerical study by utilizing different fin structures within the reactor is conducted. From the analysis, the efficient reactor configuration is determined, and further design optimization studies based on the structural and operational parameters are discussed. Moreover, the scale-up methodology is extensively discussed, and the performance of the proposed reactor design is compared with the recent studies available in the literature. Based on the analysis of the present study, the following primary outcomes are stated below:

- Five reactor configurations are numerically analyzed and their outcomes are compared. The DFMTR model shows remarkably higher performance by saving the time required to store and release hydrogen. This reactor saved 38% and 31% time to store and release 90% of hydrogen compared to the baseline reactor.
- The DFMTR absorbs maximum storage capacity i.e., 1.37 wt.% of hydrogen in 795 s, which is 46 %, 30 %, 28 %, and 27 % faster than the SMTR, SFMTR, TFMTR, and LFMTR models.
- The DFMTR discharges all stored hydrogen in 1060 s, whereas the SMTR, SFMTR, TFMTR, and LFMTR take 2245 s, 1622 s, 1494 s, and 1439 s, respectively.
- The effect of fin numbers (axial pitch) and its thickness are numerically analyzed and optimized, and the optimal values are found to be 94 fins and 0.5 mm thick, respectively.
- The effect of external jacket also analyzed and found the use of external jacket saved 30% and 28% time to store and release 90 % hydrogen.
- The impact of inlet H₂ pressure, cooling and heating fluid temperature on the performance of DFMTR is analyzed and the optimal values found to be 20 bar, 25 °C and 70 °C, respectively.
- The DFMTR is compared with recent studies of literature and found DFMTR reactor saved 22.1%, 50%, 52.1%, and 8.7% of time to store 90% hydrogen compared with tapered finned multi tube reactor, tube bundle reactor, radial finned reactor, and pin fin reactor, respectively. Therefore, the author recommends the disc fins inside the reactor to make the reactor compact, lightweight with enhanced performance.

CRediT authorship contribution statement

Shubham Parashar: Conceptualization, Methodology, Investigation, Data curation, Writing – original draft. **P. Muthukumar:** Resources, Supervision, Funding acquisition, Writing – review & editing. **Atul Kumar Soti:** Supervision, Funding acquisition, Writing – review &

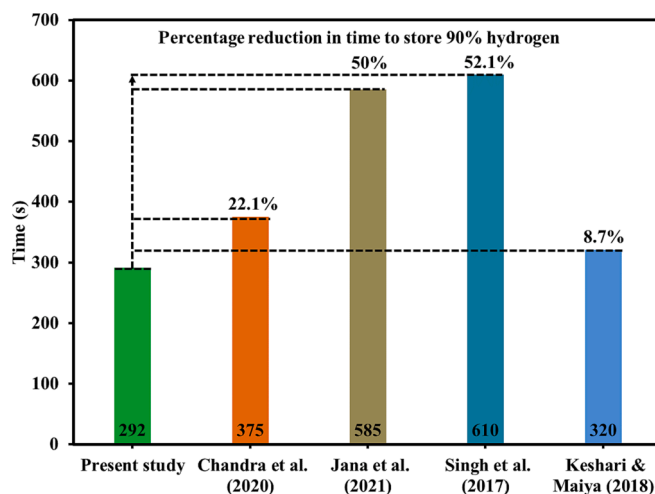


Fig. 13. Comparison in time to store 90% hydrogen of recently reported studies.

editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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